

Who Needs What? The Interaction Between Empathy Communication and Trait Empathy in a Home-Based Virtual Healthcare Robotic Agent for Older Adults

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Abstract

As healthcare interactive agents increasingly support aging in place, it is critical to understand how linguistic design interacts with older adults' empathic capacities. We examined how affective and cognitive dialogue strategies shape older adults' perceived empathy in observed video-based interactions with a virtual robotic agent and how trait empathy moderates these effects. Empathic dialogue not only directly increased perceived empathy but also interacted with traits in a complementary way. In the non-empathic baseline, trait cognitive empathy primarily moderated perceived empathy. When dialogue expressed affective empathy, cognitive trait empathy remained the key moderator, whereas when dialogue expressed cognitive empathy, affective trait empathy became dominant. This pattern suggests a systematic complementarity between agent-provided cues and human empathic capacities, indicating that targeted linguistic cues can modulate social perception and motivate designs for virtual robotic personas that adapt dialogue strategies to older adults' empathy profiles in home-based care.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **Seniors**.

Keywords

human-agent interaction; virtual healthcare robotic agents; older adults; empathy; conversational design

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1 Introduction

As populations continue to age, increasing numbers of older adults seek to manage health challenges in their homes, placing growing demands on healthcare systems [14, 30]. In response, socially assistive robots are increasingly explored as essential supports for aging in place, offering not just functional assistance but potentially companionship and daily monitoring [3, 29]. Unlike simple smart devices, these robots are expected to occupy a social role in the home. However, successful adoption depends on more than just task capability; it requires that the robot's communication feels socially appropriate and engaging [4, 7]. In home contexts, where interactions are repeated and embedded in daily routines, the nuance of a robot's dialogue style can determine whether it is embraced as a supportive partner or rejected as a mere machine [11]. Consequently, designing effective linguistic interaction is critical for the next generation of home-based healthcare robots.

In human interaction, empathy is a multilayered capacity supported by two complementary processes: affective empathy, which enables rapid emotional resonance with others, and cognitive empathy, which supports perspective taking and inference about others' beliefs and intentions [27, 33]. Although these components are dissociable, they often operate in tandem and can dynamically complement each other depending on context [27]. This complementarity is particularly relevant for robotic technologies, where empathy must be communicated through observable linguistic cues rather than assumed as an internal trait [17]. Prior work shows that empathic behaviors in artificial agents can increase perceived empathy and improve relational outcomes such as trust and satisfaction [1, 13]. However, prior studies frequently evaluate perceived empathy as a global user-facing impression, often failing to distinguish between affective and cognitive mechanisms [26]. Furthermore, it remains underexplored how these distinct dialogue strategies interact with individual differences in older adults.

In this study, we investigate how older adults attribute empathy to a virtual robotic agent in observed interactions when it expresses



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empathy through affective versus cognitive dialogue cues, and how these effects depend on older adults' trait empathy. We address the following research questions: How do affective vs. cognitive empathic dialogue cues shape older adults' perceived empathy relative to a non-empathic baseline? How do older adults' trait affective and cognitive empathy differentially moderate these effects? Our findings contribute empirical evidence on mechanism-by-trait interactions in empathy attribution, providing practical guidance for designing empathy as configurable dialogue strategies for home-based healthcare robots.

2 Background

2.1 Interactive Healthcare Agents for Older Adults

Interactive healthcare agents, ranging from voice-based assistants to socially assistive robots, are increasingly explored as essential supports for aging in place [30]. While current voice-based assistants offer convenient, hands-free access to information [23], socially assistive robots extend this utility by providing companionship and daily monitoring [29]. However, adoption relies on more than just functional utility; recent pilots suggest that for these systems to be accepted in domestic environments, they must offer a sense of social presence and engaging conversation [1, 6]. Consequently, the design challenge has shifted beyond simply asking what these agents do to understanding how their conversational behavior shapes the user experience in emotionally charged health contexts.

2.2 The Duality of Empathy: Affective vs. Cognitive

To bridge the gap between machine functionality and human connection, researchers increasingly focus on empathy. Psychological theory posits that empathy is not a monolith but a multi-component construct combining affective processes, such as sharing or mirroring emotions, and cognitive processes, involving perspective-taking and reasoning [31, 33]. While humans fluidly blend these mechanisms [27], robotic agents must be explicitly designed to express them through dialogue. Prior work in Human-Robot Interaction (HRI) indicates that explicit empathic dialogue, whether focusing on emotional resonance or intellectual understanding, can significantly enhance user trust and satisfaction compared to neutral interactions [13].

However, current research often treats empathy as a broad, uni-dimensional feature, failing to disentangle how distinct communication styles specifically impact older adults' perceptions of cognitive versus affective empathy [22, 26]. It remains unclear whether specific linguistic cues, such as "feeling" versus "understanding," successfully target their intended empathetic dimension or simply result in a generalized perception of warmth. Furthermore, empathy is fundamentally a dyadic process that depends not just on the sender but on the receiver [27, 33]. Individuals possess varying levels of trait empathy, meaning some older adults are naturally more emotionally responsive while others are more analytically inclined [12, 24]. A standardized dialogue strategy ignores these individual differences. This study addresses this gap by investigating how an

older adult's inherent empathy profile filters their interpretation of a robot's words, specifically examining whether their trait affective and cognitive empathy differentially moderate the effectiveness of distinct dialogue strategies.

3 Method

3.1 Participants

We recruited 390 U.S.-based older adults via CloudResearch Connect, strictly adhering to the National Institute on Aging's definition of older adulthood (age 65+) [21]. Following rigorous data cleaning, including two embedded attention checks, we excluded respondents who failed a check or did not meet eligibility criteria. The final sample consisted of 341 participants (210 female, 130 male, 1 non-binary) aged 65–84 years ($M = 69.91$; $SD = 4.33$). An *a priori* G*Power analysis [10] confirmed that this sample size provided high power ($1 - \beta = 0.95$) to detect medium effects ($f = 0.25$).

3.2 Stimuli Design and Empathy Expressions

We presented the healthcare robot as an animated virtual character to hold embodiment constant while maintaining control over verbal content. Prior work supports this decision, indicating that social perceptions of virtual agents are largely comparable to physical robots when co-presence is not a factor [16, 19]. All interactions were fully scripted. Scenarios varied in character gender, with each scenario presented once using either a male or a female actor, assigned at random. In each scenario, an older adult character expressed a specific concern, and the agent responded using one of three dialogue styles. In the Control (Neutral) condition (Fig. 1a), the agent provided only a task-oriented response. In the Affective Empathy condition (Fig. 1b), the agent expressed emotional resonance using templates from Liu et al. [17], such as "I can imagine feeling ..." or "I relate to feeling ...". In the Cognitive Empathy condition (Fig. 1c), the agent expressed intellectual understanding using theory-based structures like "It sounds like you feel ... because of ...".

To ensure ecological validity, we identified 24 representative use cases from a search of six major review articles on healthcare robotics [3, 7, 15, 25, 28, 32]. The final video stimuli were generated using OpenAI Sora to depict a domestic care setting. Spoken dialogue was synthesized using Amazon Polly, utilizing the "Salli" Neural voice for the agent to distinguish it from the human voices. Microsoft Clipchamp was used to standardize video length and timing, with nonverbal motion held constant across conditions, and three researchers independently reviewed all scripts to minimize unintended variations.

3.3 Study Design and Procedure

The study employed a between-subjects design with Empathic Communication Design (Affective vs. Cognitive vs. Control) as the primary independent variable. Participants were randomly assigned to one of three conditions. After providing informed consent, participants completed the QCAE scale [24] to assess trait empathy (Affective and Cognitive Empathy; 5-point Likert scale). Participants then viewed the 24 animated videos in a randomized order across four blocks. After each block, they rated the robot on the PETS scale [26] for perceived affective and cognitive empathy (0–100 scale). The session concluded with a demographics survey,

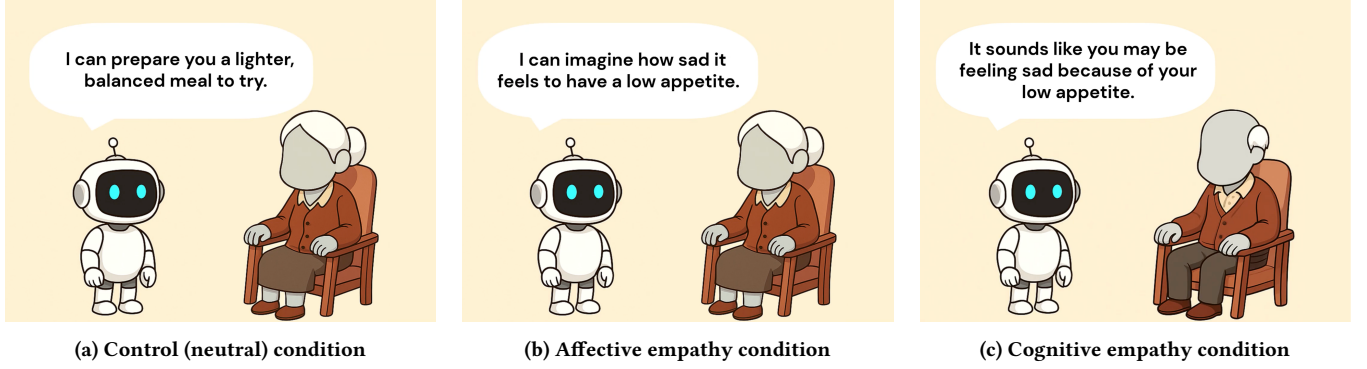


Figure 1: Dialogue design of the agent in the (a) Control (neutral), (b) Affective empathy, and (c) Cognitive empathy conditions.

and participants received \$7 compensation (total: ~ 30 min). All study procedures were approved by the Institutional Review Board at Arizona State University (STUDY00022605).

3.4 Data Analysis

We used Linear Mixed-Effects Models to examine the effects of dialogue style and trait empathy on perceived empathy. Fixed effects included the condition, trait empathy scores, and their interaction, while random intercepts accounted for repeated measures. All analyses were conducted in IBM SPSS (v29.0) with a significance threshold of $\alpha = 0.05$. Interactions were interpreted using simple slopes analysis, and post hoc comparisons applied Bonferroni corrections.

4 Results

4.1 Construct Validity and Reliability

Construct validity and reliability were examined using exploratory factor analysis (EFA). Following Reniers et al. [24], 31 QCAE items were aggregated into their established subscales to reflect the hierarchical structure. These parcels, alongside the individual items assessing perceived empathy [26], were entered into a principal component analysis with oblimin rotation. All items loaded strongly on their intended factors (> 0.60), confirming distinct constructs. Internal consistency was robust across all measures (all Cronbach's $\alpha \geq 0.83$).

4.2 Effects of Empathic Communication on Perceived Empathy

The dialogue style significantly shaped perceptions of affective empathy ($F(2, 1356.11) = 181.65, p < 0.001$) and cognitive empathy ($F(2, 1352.42) = 16.81, p < 0.001$), with the *presence* of empathy proving more impactful than the *type*. For perceived affective empathy, both the affective ($\Delta M = 25.82, p < 0.001$) and cognitive designs ($\Delta M = 22.52, p < 0.001$) significantly outperformed the control, but did not differ from each other ($p = 0.079$). Similarly, perceived cognitive empathy was higher in both the affective ($\Delta M = 7.94, p < 0.001$) and cognitive conditions ($\Delta M = 5.50, p < 0.001$) compared to the control, again with no significant difference between the two empathy types ($p = 0.254$). Overall, expressing

empathy reliably increased perceived empathy scores regardless of which strategy was used.

4.3 Individual Differences in Empathy as a Moderator of Empathy Expression Effects

4.3.1 Moderating effects on perceived affective empathy. Participants' trait empathy significantly moderated perceived affective empathy (Trait Affective: $F(2, 1347.64) = 8.02, p < 0.001$; Trait Cognitive: $F(2, 1347.64) = 8.83, p < 0.001$). As shown in Fig. 2a, the specific trait driving these perceptions varied by condition. In both the Control and Affective conditions, perception was driven primarily by participants' trait cognitive empathy (Control: $b = 16.93, p < 0.001$; Affective: $b = 7.57, p < 0.001$), while trait affective empathy was non-significant ($p \geq 0.970$). Wald tests in both cases confirmed trait cognitive empathy as the stronger driver (both $\Delta b \geq 7.50, p \leq 0.013$). However, a functional shift occurred in the Cognitive condition. Here, trait cognitive empathy lost significance ($p = 0.129$), while trait affective empathy emerged as the significant moderator ($b = 9.59, p < 0.001$). Although the Wald test was not significant ($p = 0.135$), the pattern indicates that cognitive dialogue cues may reduce reliance on user trait cognitive empathy and shifted moderation toward trait affective empathy.

4.3.2 Moderating effects on perceived cognitive empathy. A similar pattern emerged for perceived cognitive empathy (Trait Affective: $F(2, 1344.85) = 16.84, p < 0.001$; Trait Cognitive: $F(2, 1344.85) = 6.10, p = 0.002$). In the Control and Affective conditions (Fig. 2b), perceptions were driven by trait cognitive empathy (Control: $b = 14.15, p < 0.001$; Affective: $b = 6.97, p < 0.001$) rather than trait affective empathy ($p \geq 0.062$). Wald tests confirmed the dominance of the cognitive trait in both scenarios (both $\Delta b \geq 9.89, p < 0.001$). In contrast, the Cognitive Empathy condition produced a clear reversal. Trait cognitive empathy no longer significantly moderated perceptions ($p = 0.127$). Instead, trait affective empathy emerged as the robust, dominant moderator ($b = 11.50, p < 0.001$). A significant Wald test ($\Delta b = 7.90, p = 0.034$) confirmed that under cognitive dialogue design, the individual trait affective empathy exerted a significantly stronger influence than their trait cognitive empathy. Together, these findings show that perceived robot empathy depends jointly on empathic communication style and individual trait empathy.

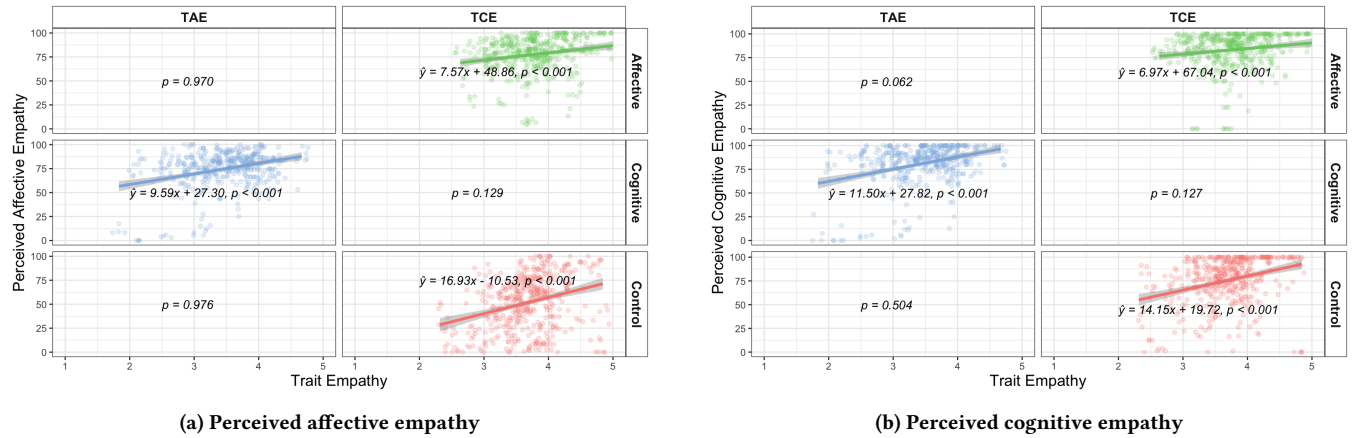


Figure 2: Moderating effects of participant trait empathy on perceived empathy. Note. TAE = trait affective empathy; TCE = trait cognitive empathy.

5 Discussion

5.1 Dialogue Design as a Social Cue for Empathy Attribution

Our first major finding demonstrates that empathic dialogue cues, whether affective or cognitive, significantly increased older adults' perceived empathy compared to a non-empathic baseline. This aligns with prior evidence that linguistic style is a primary determinant of social evaluations in health-oriented artificial agents [1, 13]. From a design perspective, this suggests that empathic language operates as a primary social signal that shapes older adults' judgments of friendliness and relational capacity. These dialogue-level insights can inform the design of other home healthcare platforms where interaction is primarily through conversation, including voice assistants and socially assistive robots [18]. This portability is further supported by evidence that older adults' engagement with voice assistants [5, 34] and healthcare robots [6, 13] is shaped by conversational experience, perceived companionship, and relational and positive information provision in health contexts. Accordingly, empathic dialogue may serve as a practical lever for improving social acceptance across home healthcare agents by making support and relational intent more legible through language.

5.2 The Complementary Role of Empathy-Related Individual Differences

While the general effect of empathy was positive, our moderation analysis reveals that the mechanism of empathy attribution relies on a complementary fit, mirroring human interaction where affective and cognitive processes operate in tandem [27, 31]. In the Control condition, older adults' perceived empathy was driven almost entirely by their own trait cognitive empathy. This aligns with the Computers Are Social Actors (CASA) paradigm [11, 20, 29], suggesting that when an agent provides few social cues, users must rely on their own inferential resources to interpret the interaction. The interaction places an uneven burden on the user to make sense of the robot's behavior through cognitive effort.

However, the introduction of specific empathic cues reconfigured this dynamic, revealing a complementary pattern [27, 31]. When the robot expressed emotion (Affective Dialogue), it continued to rely on the users' trait cognitive empathy for processing. In contrast, when the robot expressed intellectual understanding (Cognitive Dialogue), it may reduce the influence of users' trait cognitive empathy. Instead, this strategy specifically engaged users with high trait affective empathy. This suggests that successful empathy in HRI is not about matching the user, but about completing the dyad. The robot that provides logical perspective-taking appears to resonate most with users who naturally prioritize emotional connection, potentially because it offers the structure and validation they seek. Conversely, the robot that offers emotional resonance appears to require the user to cognitively process that signal. This "complementary fit" implies that agent-provided cues may redistribute older adults' own empathic work, rebalancing the social interaction.

These findings challenge the use of static, one-size-fits-all dialogue scripts. A robot programmed with a single "empathetic persona" may inevitably fail to engage a subset of older adults effectively. Instead, evidence from voice assistant [2, 5, 34] and robot deployment studies [1, 6, 9] suggests that systems must function as adaptive systems. By assessing a user's trait empathy—either during onboarding or through longitudinal interaction—a robot could dynamically toggle its linguistic strategy to provide the complementary cue the user needs. This shifts the design goal from simply "adding empathy" to configuring the type of empathy that aligns with the user's psychological needs [22], while maintaining guardrails to avoid overstating understanding or creating an illusion of empathy [4, 6, 8, 34].

5.3 Limitations and Future Work

We acknowledge several limitations. First, the study was conducted online using virtual agents. While this isolated the effects of dialogue, it may not fully capture how physical embodiment influences empathic perceptions in real-world home environments. Future work should examine how linguistic cues interact with physical

presence. Second, our sample was limited to U.S.-based participants. Given that empathy expressions are culturally coded, future studies should validate these patterns across diverse cultural demographics. Finally, future research should employ qualitative methods to better understand the subjective experience of this "complementary fit," specifically examining why users with specific trait profiles find comfort in opposing dialogue styles.

6 Conclusion

This study shows that empathic dialogue design shapes older adults' perceptions of robot empathy by redistributing empathic work between the home-based healthcare agent and the individual. Affective and cognitive dialogue strategies produce a complementary pattern in which agent-provided cues offset differences in older adults' internal empathic capacities. Framing empathic dialogue as an adaptive design resource, this work informs the development of personalized home-based voice assistants and healthcare robots that balance affective and cognitive support to enable aging in place.

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